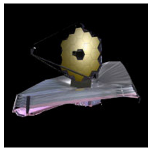

7982 - Warm Jupiters: the next step in uncovering giant planet formation and migration

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NGTS-11b				
	1	NGTS-11b SOSS	NIRISS Single-Object Slitless Spectroscopy	(1) NGTS-11
	4	NGTS-11b G395M	NIRSpec Bright Object Time Series	(1) NGTS-11

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
TOI-216c				
	2	TOI-216c SOSS	NIRISS Single-Object Slitless Spectroscopy	(2) TOI-216
	5	TOI-216c G395M	NIRSpec Bright Object Time Series	(2) TOI-216
TOI-1670c				
	3	TOI-1670c G235M	NIRSpec Bright Object Time Series	(3) TOI-1670
	6	TOI-1670c G395M	NIRSpec Bright Object Time Series	(3) TOI-1670

ABSTRACT

Understanding planetary formation and migration via their atmospheric composition is a key goal of exoplanetary characterization. However, for any individual planets, the knowledge about the protoplanetary environment is too uncertain to relate formation to current composition. Only by using samples of planets with similar dynamical properties that imply shared formation pathways, and comparing composition between these samples, can we understand how formation impacts composition. We identify a third key population with a distinct formation pathway, warm Jupiters with inner companions. Standard disc migration is disfavoured for these planets, as it would destroy the inner planetary system, implying these warm Jupiters underwent only weak migration. We would therefore expect their atmospheric composition to reflect the local inner disc. We can compare this to the populations of the BOWIE-ALIGN programme, which is testing for differences in composition between misaligned hot Jupiters believed to form via high-eccentricity migration, and aligned hot Jupiters believed to form via disc migration.

We will observe the transmission spectra of three warm Jupiters, NGTS-11b, TOI-216c, and TOI-1670c, combining them with planned observations of TOI-1130c and TOI-2525c. This will allow us to measure the C/O ratio and metallicity for these targets, which we can combine and compare to the BOWIE-ALIGN populations. We simulate our observational test, and perform 1000 random draws to demonstrate we can robustly detect a difference in composition in the face of both theoretical and observational uncertainties with 5 targets, meaning even a null result would be informative.

OBSERVING DESCRIPTION

We will obtain the near infrared transmission spectrum of three warm Jupiters, observing two transits for each target.

For each target we will observe the transit with NIRSpec in Bright Object Time Series (BOTS) mode to obtain a transmission spectrum from 2.8-5.1 microns, with a single exposure with the G395M grating. For NGTS-11b we use 37 groups/integration with 754 integrations for a total of 7.18 hours of observing time (including a 3.59 transit). For TOI-216c we use 34 groups/integration with 1257 integrations for a total of 11.03 hours of observing time (including a 5.52 hour transit). For TOI-1670c we use 8 groups/integration with 4778 integration for a total of 10.80 hours of observation (including a 5.40 hour transit). Each exposure uses SUB2048 and NRSRAPID readout.

We also observe a transit with NIRISS SOSS for NGTS-11b and TOI-216c to obtain shorter wavelength coverage, using 15 groups/integration and 296 integrations, and 14 groups/integration and 482 integrations respectively, for the same durations as their NIRSpec observations, both using SUBSTRIP256 and NISRAPID readout. For TOI-1670c, which is not observable with NIRISS due to contamination, we instead use a BOTS observation with the G235M filter, SUB2048, and NRSRAPID readout, using 4 groups/integration with 8583 integrations.

For target acquisition (TA) we use the science target for our observations of NGTS-11b and TOI-216c, using WATA mode with F110W and SUB32 for NIRSpec, and SOSSFAINT with 7 groups/integration for NIRISS. This partially saturates the central pixel (after 1 group), but there is sufficient SNR (>20) on the surrounding pixels to allow for accurate target acquisition (this exact setup, with a saturated central pixel, was done successfully for target acquisition of WASP-39 by ERS program #1366). For the observation of TOI-1670c, NIRSpec TA would have more than one saturated pixel, so we instead acquire using a nearby star 8" away, using F140X to increase the SNR.

The total charged time is 74.1 hours.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGTS-11	RA: 01 34 5.1361 (23.5214004d) Dec: -14 25 9.15 (-14.41921d) Equinox: J2000	Proper Motion RA: 11.105 mas/yr Proper Motion Dec: 13.875999999999998 mas/yr Parallax: 0.0052943" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. *SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.* *Category=Star* *Description=[Exoplanets, K stars]*				
(2)	TOI-216	RA: 04 55 55.2549 (73.9802287d) Dec: -63 15 36.31 (-63.26009d) Equinox: J2000	Proper Motion RA: -22.534 mas/yr Proper Motion Dec: -56.44900002153008 mas/yr Parallax: 0.0056152" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. *SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.* *Category=Star* *Description=[Exoplanets, K stars]*				
(3)	TOI-1670	RA: 17 16 4.1608 (259.0173367d) Dec: +72 09 40.17 (72.16116d) Equinox: J2000	Proper Motion RA: -6.007 mas/yr Proper Motion Dec: 4.923 mas/yr Parallax: 0.0060224" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. *SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.* *Category=Star* *Description=[Exoplanets, F stars]*				
(4)	TOI-1670_TA_star	RA: 17 16 5.0851 (259.0211879d) Dec: +72 09 46.34 (72.16287d) Equinox: J2000	Proper Motion RA: -0.329 mas/yr Proper Motion Dec: -14.267 mas/yr Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the 2MASS database. *link to Gaia DR3 information sheet: https://vizier.cds.unistra.fr/viz-bin/VizieR-5?-ref=VIZ670930723e468e&-out.add=.&-source=I/355/gaiadr3&-c=259.02128284098%20%2b72.16285113165,eq=ICRS,rs=2&-out.orig=o* *Object located 8" away from target star* *Category=Star* *Description=[K stars]*				

Observation

Proposal 7982, Observation 1: NGTS-11b SOSS Wed Jun 10 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRISS Single-Object Slitless Spectroscopy

Diagnostics

(NGTS-11b SOSS (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGTS-11	RA: 01 34 5.1361 (23.5214004d) Dec: -14 25 9.15 (-14.41921d) Equinox: J2000	Proper Motion RA: 11.105 mas/yr Proper Motion Dec: 13.875999999999998 mas/yr Parallax: 0.0052943" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets, K stars]

Acquisition

#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	SOSSFAINT	F480M	NISRAPID	7	1	1	0.384	222857.10

Template

Subarray	Include Short First Exposure and F277W Exposure?
SUBSTRIP256	true

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	NISRAPID	1	1	1	1	1	11.008	
2	NISRAPID	15	296	1	1	296	26025.646	222857.5
3	NISRAPID	15	10	1	1	10	879.245	222857

Special Requirements	
	Phase 0.9951935710861483 to 0.996368737079755 with period 35.455984 Days and zero-phase 2458390.70545 HJD Aperture PA Range 240 to 277 Degrees (V3 239.43873283 to 276.43873283) Time Series Observation No Parallel Attachments

Observation

Proposal 7982, Observation 4: NGTS-11b G395M

Wed Jun 10 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(NGTS-11b G395M (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGTS-11	RA: 01 34 5.1361 (23.5214004d) Dec: -14 25 9.15 (-14.41921d) Equinox: J2000	Proper Motion RA: 11.105 mas/yr Proper Motion Dec: 13.875999999999998 mas/yr Parallax: 0.0052943" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets, K stars]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	222857.7

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSRAPID	37	754	1	1	754	25859.546	222857.1

Special Requirements	
	Phase 0.9951935710861483 to 0.996368737079755 with period 35.455984 Days and zero-phase 2458390.70545 HJD Time Series Observation No Parallel Attachments

Observation

Proposal 7982, Observation 2: TOI-216c SOSS

Wed Jun 10 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRISS Single-Object Slitless Spectroscopy

Diagnostics

(TOI-216c SOSS (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	TOI-216	RA: 04 55 55.2549 (73.9802287d) Dec: -63 15 36.31 (-63.26009d) Equinox: J2000	Proper Motion RA: -22.534 mas/yr Proper Motion Dec: -56.44900002153008 mas/yr Parallax: 0.0056152" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets, K stars]

Acquisition

#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	SOSSFAINT	F480M	NISRAPID	7	1	1	0.384	222857.11

Template

Subarray	Include Short First Exposure and F277W Exposure?
SUBSTRIP256	true

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	NISRAPID	1	1	1	1	1	11.008	
2	NISRAPID	14	482	1	1	482	39731.491	222857.6
3	NISRAPID	14	10	1	1	10	824.305	222857

Special Requirements	
	Between Dates 27-DEC-2026:07:44:19 and 27-DEC-2026:08:44:19 Aperture PA Range 1 to 29 Degrees (V3 0.43873283 to 28.43873283) Aperture PA Range 59 to 80 Degrees (V3 58.43873283 to 79.43873283) Aperture PA Range 171 to 185 Degrees (V3 170.43873283 to 184.43873283) Aperture PA Range 203 to 214 Degrees (V3 202.43873283 to 213.43873283) Aperture PA Range 343 to 360 Degrees (V3 342.43873283 to 359.43873283) Time Series Observation No Parallel Attachments

Observation

Proposal 7982, Observation 5: TOI-216c G395M Wed Jun 10 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(TOI-216c G395M (Obs 5)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	TOI-216	RA: 04 55 55.2549 (73.9802287d) Dec: -63 15 36.31 (-63.26009d) Equinox: J2000	Proper Motion RA: -22.534 mas/yr Proper Motion Dec: -56.44900002153008 mas/yr Parallax: 0.0056152" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets, K stars]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	222857.8

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSRAPID	34	1257	1	1	1257	39709.233	222857.2

Special Requirements	
	Between Dates 11-AUG-2026:02:18:52 and 11-AUG-2026:03:18:52 Time Series Observation No Parallel Attachments

Observation

Proposal 7982, Observation 3: TOI-1670c G235M

Wed Jun 10 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(TOI-1670c G235M (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	TOI-1670	RA: 17 16 4.1608 (259.0173367d) Dec: +72 09 40.17 (72.16116d) Equinox: J2000	Proper Motion RA: -6.007 mas/yr Proper Motion Dec: 4.923 mas/yr Parallax: 0.0060224" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets, F stars]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	4 TOI-1670_TA_star	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	222857.12

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235M/F170LP	NRSRAPID	4	8583	1	1	8583	38885.11	222857.4

Special Requirements	
	Phase 0.9939754815596362 to 0.9949979728088493 with period 40.750145 Days and zero-phase 2459402.88333 HJD Time Series Observation No Parallel Attachments

Observation

Proposal 7982, Observation 6: TOI-1670c G395M

Wed Jun 10 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(TOI-1670c G395M (Obs 6)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	TOI-1670	RA: 17 16 4.1608 (259.0173367d) Dec: +72 09 40.17 (72.16116d) Equinox: J2000	Proper Motion RA: -6.007 mas/yr Proper Motion Dec: 4.923 mas/yr Parallax: 0.0060224" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets, F stars]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	4 TOI-1670_TA_star	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	222857.12

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSRAPID	8	4778	1	1	4778	38885.657	222857.3

Special Requirements	
	Phase 0.9939754815596362 to 0.9949979728088493 with period 40.750145 Days and zero-phase 2459402.88333 HJD Time Series Observation No Parallel Attachments